

Department of Computer Science and Engineering
Academic Year 2023– 2024(Odd Semester)

Degree, Semester & Branch: B.E, V & CSE

Course Code & Title: CS3501 & Compiler Design

Name of the Faculty member (s): Dr. I Gethzi Ahila Poornima

Innovative Practice Description

- **Unit / Topic: Unit I / Lex Tool**
- **Course Outcome: CO 1**
- **Activity Chosen: Active Learning- Learning by Teaching**

- **Justification:**

Lex and yacc are a pair of programs that help write other programs. Input to lex and yacc describes how we want our final program to work. The output is source code in the C programming language; we can compile this source code to get a program that works the way that you originally described.

- **Time Allotted for the Activity: 30 Minutes**

- **Details of the Implementation:**

Learning by teaching is a teaching approach in which students are required to acquire content and prepare lessons in order to teach it to other students. In prior, the students were informed about the topic and the event itself (exactly 5 days before) so they would have time to prepare. Two students from were chosen at random to present their topic on the board or using a power point show. Ms.Abinaya and Ms.Swathi were asked to explain about the Lex tool and their role in compilation process. Ms Swetha Sri presented the basic rules and syntax for writing basic program in Lex and she elaborated the process of executing a lex program and its output. Figure 1 shows a sample photo taken during the event when Ms.Abinaya presented her ideas on stage.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PSO1	PSO2
CO2	3	2	2	2	1

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PSO1	PSO2
	3	2	2	2	1
Justification for correlation	The students will apply the principles of computer science and	The students will evaluate problem statements and identify	The students will able to design the scanner part of a	The students will exhibit the knowledge of lexical analysis	The students will be able to use the knowledge of compiler

	engineering to design the compiler phases	modules and parameters to solve a problem.	compiler.	phase to develop software.	design phases for developing IT solutions
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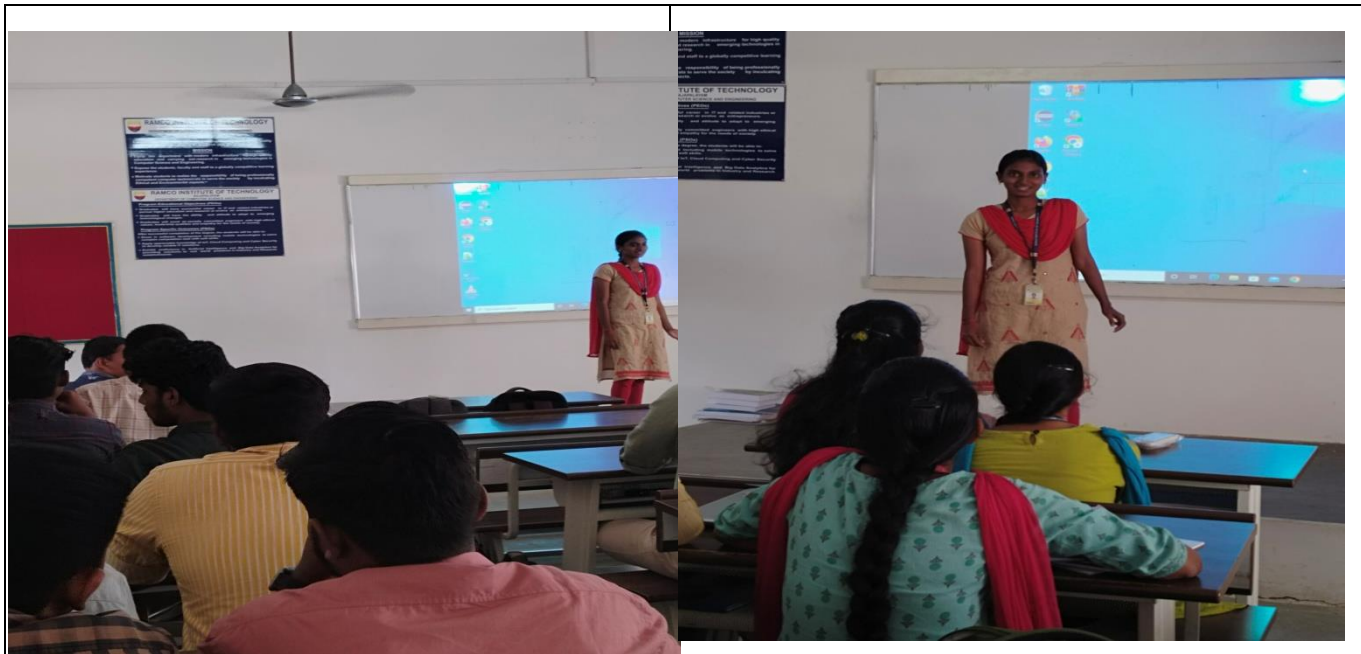


Figure 1: Photo taken during the presentation by Ms Abinaya (953621104001)

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- The practice made all of the students happy because it allowed them to thoroughly prepare for the subject.
- Since this topic is in Laboratory class, the student's involvement in this event was appreciable.

- ❖ ***Benefit of the practice:***

- They were easily able to remember the material after it was explained to them, and they were also ready for any questions that might come up throughout the session.

- ❖ ***Challenges faced in implementation:***

- A relatively small percentage of students were reluctant to speak up on stage.

Signature of Faculty Member

HOD